UNSW ZZSC9020 - Data Science Project - Task Planner (Group A)

	Tasks	Phase	Assignee	Status	LOE (D)	Priority	Description	Start Date	End Date	Assessment
1	Team Contract, roles, tooling setup and finalise Research Topic	Project Planning	All Members	Complete	1	High	Agree the working contract and decision rules, confirm roles, and stand up Teams, Planner, OneDrive and the GitHub repository.	Sep 1, 2026	Sep 3, 2026	A1A - Plan
2	Weekly Lecturer Consultation	Project Planning	All Members	In Progress	7	Medium	Book the slot, circulate an agenda to the repository 24 hours before, publish minutes within 24 hours after.	Sep 3, 2026	Oct 16, 2026	All
3	Project Plan S1: Introduction and Motivation	Project Planning	Daniel Cao	In Progress	2	Medium	Write section 1: the business problem, why medium-term demand forecasting matters, who the client is, and what the project sets out to achieve.	Sep 1, 2026	Sep 6, 2026	A1A - Plan
4	Project Plan S2: Brief Literature Review	Project Planning	Arman Hajisafi Kelly Hamilton	In Progress	4	High	Write section 2: survey existing work on medium- and long-term load forecasting, identify the methodological gap, and state how this project addresses it.	Sep 1, 2026	Sep 6, 2026	A1A - Plan
5	Project Plan S3: Methods, Software and Data Description	Project Planning	Faiza Mumtaz Swapnil Shinde	In Progress	4	High	Write section 3: the intended models & why they suit the question, software stack, and a description of the three datasets including format, size and known quality issues.	Sep 1, 2026	Sep 6, 2026	A1A - Plan
6	Project Plan S4: Activities and Schedule	Project Planning	Xavier Sun	In Progress	2	Medium	Write section 4: the activity list, the phase structure, role justification, the timeline and the Gantt chart.	Sep 1, 2026	Sep 6, 2026	A1A - Plan
7	Group Review Draft - Project Plan	Project Planning	All Members	Not Started	1	Medium	Read the four sections together, resolve overlaps and contradictions, check the three-page limit, and submit.	Sep 6, 2026	Sep 6, 2026	A1A - Plan
8	Complete Individual Implementation Checklist	Project Planning	All Members	In Progress	1	High	Assessment 1B: Each member's checklist by 5 pm Tuesday of Week 2.	Sep 5, 2026	Sep 8, 2026	A1B - Checklist
9	Complete Indivdiual Weekly Self-Reflection	Reporting & Communication	All Members	In Progress	4	Medium	An individual task: write a 500-word summary reflective portfolio, based on your weekly teamwork reflections for Weeks 1-4.	Aug 31, 2026	Sep 29, 2026	A2 - Reflection
10	Ingest validate and clean datasets	Data Engineering & Preparation	Daniel Cao	Not Started	2	Medium	Load the three CSVs and confirm columns, types, units, date coverage and row counts against the course brief, check duplicates, and resample temperature onto the 30-minute demand grid.	Sep 7, 2026	Sep 9, 2026	A3 - Report
11	Build feature set and design rolling-origin splits	Data Engineering & Preparation	Arman Hajisafi	Not Started	3	Medium	Calendar and school-holiday flags, Fourier seasonality terms, a long-run trend and cooling/heating degree days on a fixed 18 C base, which activity 13 then validates rather than supplies; set the train and test origins for each horizon, then review every feature for information that would not exist at forecast time.	Sep 9, 2026	Sep 12, 2026	A3 - Report
12	Load profiles and seasonality	Exploratory Data Analysis	Faiza Mumtaz	Not Started	2	Medium	Demand by hour, weekday, month and year; STL decomposition and autocorrelation to size the seasonal structure the models have to capture.	Sep 9, 2026	Sep 11, 2026	A3 - Report
13	Temperature–demand response and trend and structural breaks	Exploratory Data Analysis	Swapnil Shinde	Not Started	3	Medium	Fit the U-shaped response curve, estimate the heating and cooling thresholds, and quantify how many megawatts each additional degree costs. Quantify the daytime decline driven by rooftop solar across 2010–2021 (isolate the 2020 COVID-19 period).	Sep 9, 2026	Sep 12, 2026	A3 - Report
14	Baseline Modelling	Modelling	Xavier Sun	Not Started	3	High	Seasonal naive (same period last year) and a climatology profile averaged over prior years. These set the bar every other model has to clear, and every later result is reported relative to them.	Sep 12, 2026	Sep 15, 2026	A3 - Report
15	Statistical models - SARIMAX	Modelling	Swapnil Shinde	Not Started	3	High	Fit SARIMAX with temperature as an exogenous input, with ETS and Prophet as comparators; check residual diagnostics and report the fitted seasonal structure.	Sep 12, 2026	Sep 15, 2026	A3 - Report
16	ML Modelling - Random Forest	Modelling	Daniel Cao	Not Started	3	High	Train a random forest on the horizon-safe feature set, tuned only on validation origins so the test origins stay untouched; record feature importances for the discussion.	Sep 12, 2026	Sep 15, 2026	A3 - Report
17	ML Modelling - XGBoost	Modelling	Faiza Mumtaz	Not Started	3	High	Train a gradient-boosted tree ensemble with XGBoost on the same features and splits; tune depth, learning rate and regularisation on validation origins only.	Sep 12, 2026	Sep 15, 2026	A3 - Report
18	ML Modelling - LightGBM	Modelling	Arman Hajisafi	Not Started	3	High	Train a LightGBM ensemble on the same features and splits, and compare its training cost and accuracy against XGBoost, which uses the same algorithm family with a different split-finding strategy.	Sep 12, 2026	Sep 15, 2026	A3 - Report
19	ML Modelling - LSTM	Modelling	Kelly Hamilton	Not Started	3	High	Train an LSTM sequence model on the same features and splits, and report whether the extra capacity earns its cost at these horizons.	Sep 12, 2026	Sep 15, 2026	A3 - Report
20	Model evaluation & validation	Evaluation & Validation	All Members	Not Started	2	Medium	Each member puts their own model through the shared harness: fit at every rolling origin, predict the horizon, collect MAE, RMSE and MAPE. Then break the error down by season, temperature band and demand level, test summer peaks and extreme-heat days.	Sep 16, 2026	Sep 18, 2026	A3 - Report
21	Model selection and freeze	Evaluation & Validation	All Members	Not Started	1	Medium	Compare the six models on the common metrics, choose the best per horizon, record the decision and the evidence behind it, and stop experimenting.	Sep 19, 2026	Sep 20, 2026	A3 - Report
22	Selected model improvement	Evaluation & Validation	Model Winner	Not Started	2	Medium	The member whose model wins takes one focused round of tuning or feature refinement, with the rest of the team reviewing. The owner is decided at Task 21 (Model Selection).	Sep 21, 2026	Sep 23, 2026	
23	Report skeleton and section owners, and draft the report chapters	Reporting & Communication	All Members	Not Started	1	Medium	Agree headings, page budget and a named owner for each of the seven report chapters, then start drafting against the frozen results.	Sep 21, 2026	Sep 22, 2026	A3 - Report
24	Group Report: Introduction + Conclusion and Further Issues + Organising Reference & Appendix	Reporting & Communication	Arman Hajisafi	Not Started	5	Medium	• Chapter 1: Introduction • Chapter 7: Conclusion and Further Issues • Reference + Appendix: Sorting & Organising	Sep 23, 2026	Oct 7, 2026	A3 - Report
25	Group Report: Literature Review + Discussion	Reporting & Communication	Kelly Hamilton	Not Started	5	High	• Chapter 2: Literature Review • Chapter 6: Discussion	Sep 23, 2026	Sep 29, 2026	A3 - Report
26	Group Report: Material and Methods: 3.1, 3.2 and 3.3	Reporting & Communication	Swapnil Shinde	Not Started	5	Medium	• Chapter 3: • 3.1 Software • 3.2 Description of the Data • 3.3 Pre-processing Steps	Sep 23, 2026	Sep 29, 2026	A3 - Report
27	Group Report: Material and Methods: 3.4, 3.5 and 3.6	Reporting & Communication	Daniel Cao	Not Started	5	High	• Chapter 3: • 3.4 Data Cleaning • 3.5 Assumptions • 3.6 Modelling Methods	Sep 23, 2026	Sep 29, 2026	A3 - Report
28	Group Report: Exploratory Data Analysis	Reporting & Communication	Faiza Mumtaz	Not Started	5	Medium	Chapter 4, written from the EDA notebooks, with every figure regenerated from the cleaned data rather than pasted.	Sep 23, 2026	Sep 29, 2026	A3 - Report
29	Group Report: Analysis and Results	Reporting & Communication	Xavier Sun	Not Started	5	High	Chapter 5: the model comparison, the stratified error tables, and what they say about which model wins under which conditions.	Sep 26, 2026	Oct 2, 2026	A3 - Report
30	Internal review and reproducibility check, and final submission	Reporting & Communication	All Members	Not Started	4	Medium	Clone the repository fresh, re-run the pipeline end to end, confirm every figure and number in the report regenerates, then submit.	Sep 29, 2026	Oct 7, 2026	A3 - Report
31	Video storyboard and slides	Reporting & Communication	All Members	Not Started	2	Low	Storyboard fifteen minutes for a non-technical audience and build one shared deck rather than six.	Oct 7, 2026	Oct 10, 2026	A4 - Presentation
32	Record, edit and submit the video	Reporting & Communication	Xavier Sun	Not Started	2	Medium	Each member records their own section; edit the pieces together and submit by 10 pm Tuesday of Week 7.	Oct 10, 2026	Oct 13, 2026	A4 - Presentation
33	Project coordination	Project Planning	Xavier Sun	In Progress	4	Medium	Maintain the task board and Gantt, keep the risk log current, write the weekly agendas and minutes, and enforce the branch-and-review convention on the repository.	Aug 31, 2026	Oct 13, 2026	All

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UNSW ZZSC9020 - Data Science Project - Task Planner Timeline (Group A)